

MA 222: Applied Linear programming

Course Code	MA 222
Course Name	Applied Linear programming
L-T-P-C	3-1-0-4
Prerequisites	Linear algebra
Intended for	B.Tech. Mathematics & Computing 2nd Year
Distribution	Core
Mutual Exclusion	MA 515
Approval	59th BoA (Apr 2025)

Course Contents

- **Linear Programming Fundamentals:** Optimization Basics and Origin of LP Problems, Systems of Linear Inequalities, Convex Sets and Feasible Regions. [4 Lectures]
- **Computational Aspects of Linear Programming:** Graphical Method, Simplex Method, Big-M, Dual Problem, Dual Simplex Method, Primal-Dual method. [14 Lectures]
- **Alternative approaches of Linear Programming:** Karmarkar's Algorithm, Transportation problems, Assignment problems. [10 Lectures]
- **Integer Programming:** Introduction to Integer Programming, Gomoly's cutting plane algorithm, Branch and Bound Method, Solving Zero-one problem. [6 Lectures]
- **Game Theory:** Basic Concept and Terminologies, Minimax (Maximin) criterion, Saddle point, Two-person Zero-sum Game, and Game with Pure and Mixed Strategies, Dominance Principle, Arithmetic Method, and Graphical Method for Solving $(2 \times n)$ Game, Graphical Method for Solving $(m \times 2)$ Game and Linear Programming Methods for Rectangular Games. [4 Lectures]

Text Books

1. Luenberger and Ye., **Linear and Nonlinear Programming**, Springer, 2008.
2. Elliott Mendelson and Daniel Zwillinger, **Introducing Game Theory and its Applications**, 2nd Edition, Taylor & Francis, 2004.

References

1. M.S. Bazaraa, J.J. Jarvis and H.D. Sherali, **Linear Programming and Network Flows**, 2nd Edition, Wiley & Sons, 1990.
2. I. Griva, S.G. Nash, A. Sofer, **Linear and Nonlinear Optimization**, SIAM, 2009.
3. Nocedal and Wright, **Numerical Optimization**, Springer, 2006.
4. Murty, Katta G., ed., **Case Studies in Operations Research: Applications of Optimal Decision Making**, Vol. 212. Springer 2014.
5. Articles as chosen by the instructor.